

ORIGINAL ARTICLE

Multidrug resistant staphylococci isolated from pigs with exudative epidermitis in North eastern Region of India

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Significance and impact of the study: Exudative epidermatitis (EE) or greasy pig disease (GPD) is a contagious and endemic disease of pigs worldwide. The disease is not reported from India. The present study reported an outbreak of EE in Mizoram adjoining to the southern border of Myanmar. The pathogens were isolated, identified and characterized for their toxigenic potential. Majority of the isolates were multidrug resistant including methicillin and vancomycin. This is the first ever report of GPD in India and the pathogens might be transmitted through illegal migration of pigs from Myanmar to India.

Keywords

exudative epidermatitis, India, multidrug resistance, vancomycin.

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Abstract

Exudative epidermatitis or greasy pig disease (GPD) is a contagious disease of pig and endemic worldwide caused by toxigenic strains under genus *Staphylococcus*. The present study reported an outbreak of GPD in Champhai district of Mizoram adjoining to the southern border of Myanmar. A total of 60 samples were collected from 22 clinically affected animals and processed for isolation and identification of *Staphylococcus* spp. All the isolates were subjected to antimicrobial sensitivity assay, biofilm production assay and detection of virulence genes, biofilm genes and *mec* genes followed by cloning and sequencing for phylogenetic analysis. A total of 44 staphylococci belonged to four species (*S. sciuri*, *S. aureus*, *S. lentus*, and *S. hyicus*) were isolated. Majority of the isolates were multidrug resistant with maximum resistance against ampicillin, penicillin including vancomycin. None of the *S. hyicus* isolates was methicillin resistant (MRSH) but 66-67% isolates were MRSA. By PCR, *mecA* gene was detected in *S. aureus* ($n = 2$), *S. sciuri* ($n = 4$) and *S. lentus* ($n = 3$). Biofilm associated gene *icaD* was detected in *S. aureus* ($n = 3$), *S. sciuri* ($n = 5$), *S. hyicus* ($n = 4$) and *S. lentus* ($n = 6$). The exfoliative toxin genes (*ehxB*, *shetA* and *tsst1*) were detected in *S. hyicus* ($n = 3$) and *S. aureus* ($n = 1$) isolates. All the isolates were closely related with the isolates from pigs of China, Germany, Japan and USA. The pathogens might be transmitted through illegal migration of pigs from Myanmar to India.

Introduction

Exudative epidermitis (EE), which is also commonly termed as “greasy pig disease (GPD)” is a generalized or localized disease of pigs with characteristic exfoliation, sebaceous exudation and sometimes formation of crusts on the body surface (Park *et al.* 2013a). GPD is most commonly associated with toxigenic strains of *Staphylococcus hyicus*. In addition, the similar kind of disease

conditions caused by toxigenic strains of *Staphylococcus aureus* and *Staphylococcus chromogenes* are also reported (Werckenthin *et al.* 2001; Park *et al.* 2013b). Pathogenic strains of staphylococci used to produce exfoliative toxins, which can digest desmoglein-1 in the epidermis of porcine skin (Nishifuji *et al.* 2008). Due to cleavage of the desmosomal cadherin-type cell–cell adhesion molecule the cutaneous defense barrier disrupts, which facilitates bacterial invasion (Nishifuji *et al.* 2008). The mature pigs

generally develop resistance to EE with age but they may remain as carriers and can spread the organisms to the healthy herds. Affected pigs with skin lesions are treatable with antibiotics but antibiotic resistant staphylococci associated with EE are frequently developed, which makes the infection highly difficult to treat (Park *et al.* 2013a; Vailancourt *et al.* 2018).

The disease occurs worldwide and is considered as endemic problem on many pig farms. Occasionally, it causes outbreaks involving large numbers of pigs or piglets. The North eastern part of India, particularly the Mizoram state (23.1645°N, 92.9376°E) is attached with southern international border of Myanmar. In recent past, porcine reproductive and respiratory diseases syndrome (PRRS) outbreak was first time reported in Mizoram and the virus was believed to be entered from Myanmar through illegal movement of pigs (Rajkhowa *et al.* 2016). Illegal movement of pigs and piglets from Myanmar to Mizoram is frequent due to insufficient availability of piglets in Mizoram. In a recent study, we have first time reported the presence of pathogenic *Actinobacillus pleuropneumoniae* and *Haemophilus parasuis* in the pigs illegally transported from Myanmar to Champhai district of Mizoram (unpublished data). Detection of such emerging pathogens in the pigs of Champhai district of Mizoram prompted us to screen the pigs of same place for EE and to characterize the pathogens associated with EE. The present study was undertaken on isolation, identification, antimicrobial profile and molecular characterization of staphylococci associated with EE in Champhai district of Mizoram adjoining to Myanmar.

Results and Discussion

A total of 44 staphylococci belonged to four different species (*S. sciuri*, *S. aureus*, *S. lentus* and *S. hyicus*) were isolated and identified as confirmed by the BD Phoenix™ automated bacterial identification system and 16S rRNA based PCR assay. *Staphylococcus sciuri* and *S. lentus* were recovered from all three sources, whereas *S. aureus* and *S. hyicus* were recovered from clinical skin lesions only.

Staphylococcus sciuri and *S. lentus* were also recovered from infected skin lesions (Table 1).

As depicted in Table 2, phenotypically, majority of the isolates were recorded as multidrug resistance (MDR) as they were recorded as resistant to three or more than three classes of antibiotics. Maximum resistance was observed against ampicillin and penicillin. Cefotaxime along with the methicillin was used as surrogate antibiotic to detect the methicillin resistance staphylococci and both have exhibited equivalent resistance profile for all the isolates. Phenotypically, none of the isolates of *S. hyicus* was methicillin resistant (MRSH) but 66.67% isolates were recorded as methicillin resistant *S. aureus* (MRSA). Most of the staphylococci including all the *S. aureus* isolates were determined as biofilm producer by microtiter plate assay.

By PCR, *mecA* gene was detected in *S. aureus* (*n* = 2), *S. sciuri* (*n* = 4) and *S. lentus* (*n* = 3). None of the isolates were positive for *mecB* or *mecC* genes. Only *icaD* gene associated with biofilm production was detected in *S. aureus* (*n* = 3), *S. sciuri* (*n* = 5), *S. hyicus* (*n* = 4) and *S. lentus* (*n* = 6). *icaA* gene could not be detected in any of the isolates under the study. The exfoliative toxin genes (*ehxB*, *shetA* and *tsst1*) were detected in *S. hyicus* (*n* = 3) and *S. aureus* (*n* = 1) isolates only.

The nucleotide sequence analysis of exfoliative toxin genes of the isolates under the present study revealed a close relationship among the isolates from pigs of Mizoram and also with the sequences retrieved from China, Germany, Japan and USA (data not shown). The deduced amino acid sequence for the exfoliative toxin genes of *S. hyicus* revealed that most of the central part of the toxin is in close proximity to the three amino acids, which is considered to be the active site of ETA, i.e. His-72, Asp-120, Ser-195. This catalytic triad of serine protease enzyme indicates the common mode of action of the exfoliative toxins.

The EE or GPD is a communicable skin infection of pigs caused by an array of pathogenic species under the genus *Staphylococcus* (Park *et al.* 2013; Leekitcharoenphona *et al.* 2016; Lu *et al.* 2016). The predominant

Table 1 Details of isolation and biofilm production of *Staphylococcus* spp. from pigs with clinical signs of GPD

Staphylococci isolated	Nasal swab (<i>n</i> = 22)	Skin scrap/swab (<i>n</i> = 22)	Vaginal swab (<i>n</i> = 16)	Total (<i>n</i> = 60)	Biofilm production
<i>Staphylococcus sciuri</i>	10 (45.45)	9 (40.91)	3 (18.75)	22	18 (81.82)
<i>Staphylococcus aureus</i>	—	3 (13.64)	—	3	3 (100)
<i>Staphylococcus lentus</i>	9 (40.91)	2 (9.09)	3 (18.75)	14	11 (78.58)
<i>Staphylococcus hyicus</i>	—	5 (22.73)	—	5	4 (80.0)
Total	19	19	6	44	36 (81.82)

Figures in parenthesis are percentage value

Table 2 Antimicrobial resistance profile of *Staphylococcus* spp. isolated from pigs with clinical signs of GPD

Staphylococci isolated	Ampicillin (10 µg)	Cefoxitin (30 µg)	Erythromycin (15 µg)	Gentamicin (10 µg)	Methicillin (5 µg)	Penicillin (10 U)	Streptomycin (10 µg)	Tetracycline (30 µg)	Vancomycin (MIC, µg ml ⁻¹)
<i>Staphylococcus sciuri</i> (n = 22)	90.91%	27.27%	45.45%	45.45%	27.27%	100%	40.91%	31.82%	≤4: 13 (59.09%) 8–16: 7 (31.82%) ≥32: 2 (9.09%)
<i>Staphylococcus aureus</i> (n = 3)	100%	66.67%	66.67%	33.33%	66.67%	66.67%	66.67%	33.33%	≤2: 1 (33.33%) 4–8: 2 (66.67%) ≥16: 0%
<i>S. lentus</i> (n = 14)	57.14%	71.43%	35.71%	35.71%	71.43%	71.43%	42.86%	28.57%	≤4: 8 (57.14%) 8–16: 6 (42.86%) ≥32: 0%
<i>Staphylococcus hyicus</i> (n = 5)	60.00%	0.0%	40.00%	100%	0.0%	60.00%	80.00%	0.0%	≤4: 2 (40.0%) 8–16: 2 (40.0%) ≥32: 1 (20.0%)

species associated with the disease is *S. hyicus* but many other species, viz., *S. aureus*, *S. sciuri* and *S. chromogenes* were also reported by different workers from time to time (Ole Andresen et al. 2005; Lu et al. 2016). In the present study, beside the three reported species (*S. aureus*, *S. sciuri* and *S. hyicus*), *S. lentus* was also isolated from clinical cases of EE in pigs of Mizoram, India. Isolation of unconventional species of staphylococci from clinical cases of EE was also reported by Park et al. (2013a), where in addition to *S. hyicus* and *S. aureus* the other species including *S. chromogenes*, *S. pseudointermedius*, *S. arlettae* and *S. cohnii* subsp. *urealyticus* were recorded. In the present study, all the isolates were not confirmed as pathogenic through detection of selected virulence associated genes. The exfoliative toxin genes were detected in *S. hyicus* and *S. aureus* indicates their direct role in development of the disease. Detection of *S. sciuri* and *S. lentus* from the clinical cases of GPD in pigs without any detectable toxigenic genes should not be neglected as non-pathogenic. Because Lu et al. (2016) from China reported clinical case of EE by nontoxigenic *S. sciuri*, which might be supportive information that the disease may also develop in susceptible animals without evidence of its toxigenicity. Few other factors might be involved in colonization of nontoxigenic strains of staphylococci and virulence in pigs including interactions with other skin microbiota as hypothesized for *S. aureus* (Yan et al. 2013) and host immune status (Leekitcharoenphona et al. 2016).

Treatment failure in EE cases of pigs is frequently reported, which may be due to the presence of MDR staphylococci in the pig's environment (Park et al. 2013a). Resistance against the β-lactam antibiotics like ampicillin, penicillin G as well as cephalosporins is very common probably due to use of penicillin as the primary drug of choice in the treatment of EE. And in most of the occasions, treatment of EE follows in the absence

of any cultural and susceptibility testing (Park et al. 2013a). In the present study, majority of the isolates exhibited high level of resistance against most of the antibiotics including penicillin, ampicillin and vancomycin. Methicillin resistance was predominant in case of *S. lentus* (MRSL), *S. aureus* (MRSA) and *S. sciuri* (MRSS) but none of the *S. hyicus* isolates were methicillin resistant (MRSH). In addition, *mecA* gene was also detected in *S. aureus* (n = 2), *S. sciuri* (n = 4) and *S. lentus* (n = 3). Demonstration of MRSA in the pig population as well as other domestic animals was reported elsewhere due to its potential public health concern (Park et al. 2013). It is also indicated that MRSA has acquired the *mecA* from coagulase negative staphylococci, which are common commensals (Sasaki et al. 2010; Tsubakishita et al. 2010) and the methicillin resistance in other pathogenic species of staphylococci might be also acquired in the similar manner. In addition, possible transmission of *mecA* gene between *S. aureus* and *S. hyicus* in pig farms due to application of beta lactam antibiotics as prophylactic also mentioned earlier (Park et al. 2013). The findings of present study may also incorporate another possibility for transmission of *mecA* gene between *S. aureus*, *S. sciuri* and *S. lentus* in the similar fashion, although we could not record the treatment practice applied to the affected pigs adopted by the veterinarians. *mecA* gene demonstrated in non-toxigenic but pathogenic *S. sciuri* NJ1306 strain in China, which was a potential source of transmission of the gene to other pathogenic species of staphylococci (Lu et al. 2016). Vancomycin is one of the drugs of choice for treatment of MRSA associated infections, although no longer as a first choice, in humans in general and animals in particular. In a recent study, vancomycin resistant *S. aureus* was reported from swine population of entire South Italy (Pirolo et al. 2019). But vancomycin sensitive *S. aureus* from bovine and swine population was reported from Punjab, India (Zahera et al. 2017). A

vancomycin-intermediate livestock-associated methicillin-resistant *S. aureus* (ST398/t9538) is reported from swine in Brazil (Moreno *et al.* 2016). All the observations are indicative of circulation of vancomycin resistant staphylococci in the environment of human and animals, irrespective of using the drug as treatment option in animals. In the present study, detection of vancomycin resistant pathogenic staphylococci from infected pigs is a matter of concern. Vancomycin resistance is also predominantly detected in *S. sciuri*, *S. hyicus* and *S. lentus* isolates, which warrants a further detailed study of the isolates to understand the potential public health threat.

It is now an established fact that treating staphylococcal infections becoming more difficult due to rapid development of resistance to multiple antimicrobial agents. The situation is further worsening due to biofilm production, which facilitates their survival in presence of antimicrobials (Bhattacharya *et al.* 2018). The earlier report also established the fact that the biofilm producing strains of staphylococci exhibits higher tendency to become MDR including methicillin resistance compared to non-biofilm producing strains (Neopane *et al.* 2018). In the present study, all the *S. aureus* and majority of the *S. sciuri*, *S. hyicus* and *S. lentus* isolates were recorded as biofilm producer (Table 1). And *icaD* was detected in *S. aureus* ($n = 3$), *S. sciuri* ($n = 5$), *S. hyicus* ($n = 4$) and *S. lentus* ($n = 6$) isolates as biofilm associated gene. *icaA* and *icaD* genes were detected in all the *S. aureus* isolates from pigs in China (Zhang *et al.* 2018), however, our result is in corroboration with the work reported by Darwish and Asfour (2013), in which they reported a higher level of *icaD* gene in comparison with *icaA* gene. In earlier studies from our laboratory, higher incidence of *icaD* over the *icaA* gene was also reported in biofilm producing *S. aureus* isolated from cattle milk of Mizoram (unpublished data). The persistence of staphylococci is attributed to several virulence factors including the biofilm formation capacity (Simojoki *et al.* 2012). Staphylococcal biofilm formation is associated with intracellular adhesion locus (*icaABCD*), encoding the proteins required for synthesis of poly-*N*-acetyl glucosamine (PNAG), biofilm-associated protein (bap), accessory gene regulator locus (*agr*) and *IS257* (Cucarella *et al.* 2004). The *icaA* and *icaD* play the significant role in synthesis of PNAG oligomers to generate staphylococcal biofilms (O'Gara 2007).

Exfoliative toxin production is the most important virulence factor associated with EE condition in pigs. Six different gene loci have been reported among *S. hyicus* responsible for production and exfoliative toxins (Onuma *et al.* 2011). In the present study, the exfoliative toxin genes (*ehxB*, *shetA* and *tst1*) were detected in *S. hyicus* ($n = 3$) and *S. aureus* ($n = 1$) isolates only. A variable range of exfoliative toxin genes (*ehx*) was reported in the

S. hyicus isolates from pigs of eight countries and correlated with the toxin production and clinical conditions of the pig (Andresen 2005). Detection of *ehx* genes in the *S. hyicus* isolates is almost confirmatory of being a pathogenic strain, but many other staphylococci (*S. sciuri*, *S. lentus*, *S. chromogens*) were also isolated from pigs with clinical signs of EE. In the present study, *ehx* genes were not detected in any of the *S. sciuri*, *S. lentus* and *S. aureus* isolates but the isolates were biofilm producer and significantly resistant to multiple antimicrobial agents including vancomycin. Although, the well-known exfoliative toxin genes of *S. hyicus* is considered as major determinants to differentiate between the toxigenic and non-toxigenic strains, some other factors may also be responsible in colonization and virulence of pathogenic staphylococci in pigs (Leekitcharoenphon *et al.* 2016). Nucleotide sequence analysis of partial sequence of *exhB*, *shetA* and *tst-1* genes revealed close relationship with the nucleotide sequences of similar genes retrieved from China, Japan and USA. Sequence data of *ehx* or other relevant genes associated with exfoliative dermatitis in pigs in Myanmar could not be retrieved to establish the fact that the disease might have transmitted through porous Indo-Myanmar border. But the earlier evidence of transmission of highly pathogenic PRRSV in pigs from Myanmar to Mizoram, India is available (Rajkhowa *et al.* 2016). Illegal movement of pigs and piglets from Myanmar to Mizoram, India through porous international border is frequent and is considered as one of the major channels of transboundary transmission of animal pathogens (Rajkhowa *et al.* 2016). Deduce amino acid sequence analysis revealed a close similarity of the amino acid sequence pattern of all the *ehx* genes sequences of present study. The central part of the toxin is in close proximity to the three amino acids, which is considered to be the active site of ETA i.e. His-72, Asp-120, Ser-195. Presence of this catalytic triad of serine protease enzyme in *exhB* (MF399267) indicates the common mode of action of the exfoliative toxin.

It is reported that nearly 30% of the total pig population of India is maintained in the North-eastern states including Mizoram. The entire region is facing deficit to become the self-sufficient in pork production, which results in influx of pigs in this region from neighbouring states of India as well as through porous international border of Myanmar (Rajkhowa *et al.* 2016). The first outbreak of highly pathogenic PRRS (HP-PRRS) was reported for the first time in Myanmar during 2011 and subsequently in Mizoram in 2013, which is the first report of PRRS in India. Recently, from our laboratory, two important bacterial pathogens, *H. parasuis* and *A. pleuropneumoniae* were detected in the pigs of Mizoram procured illegally from Myanmar (unpublished data). *Haemophilus parasuis* and *A. pleuropneumoniae* in pigs are

also not yet reported in India. Detection of multidrug resistant pathogenic staphylococci associated with exfoliative dermatitis in pigs illegally migrated from Myanmar to Champhai district of Mizoram indicates its trans-boundary transmission. Majority of the isolates were confirmed as vancomycin resistant and also have potential public health concern.

Materials and Methods

Collection of samples

Samples were collected from Champhai district of Mizoram located at southern international border of Myanmar. A total of 22 pigs were recorded with suspected clinical signs of GPD and were subjected to collection of samples including scraps/swabs from affected skin (22), nasal swabs (22) and vaginal swab of affected sows (16). Samples were collected randomly from lesions using cotton-tipped swabs after applying 1 ml 0.9% sodium chloride on the lesions. All the samples were kept in clean sterile containers and transported to the laboratory.

Isolation and identification of bacteria

All the samples were processed in the Bacteriology Laboratory of Department of Veterinary Microbiology, Central Agricultural University as per standard method for isolation and identification of staphylococci (Quinn *et al.* 2004). All the suspected bacterial colonies were subjected to Grams staining and screened by BD Phoenix™ automated bacterial identification system (Becton, Dickinson and Company, Haryana, India) for determination of species as per the instruction of the manufacturer. The pure bacterial isolates were further confirmed by *Staphylococcus* genus specific 16S rRNA based PCR assay (Zhang *et al.* 2004).

Antimicrobial susceptibility assay

All the bacterial isolates were subjected to antimicrobial susceptibility assay by disc diffusion method as per the recommendation of Clinical Laboratory Standard Institute (CLSI 2018). The assay was performed on Mueller–Hinton agar (HiMedia, Mumbai, India) plate using nine selected commercially available antibiotic discs: ampicillin (10 µg), cefoxitin (30 µg), erythromycin (15 µg), gentamicin (10 µg), methicillin (5 µg), penicillin (10 units), streptomycin (10 µg), tetracycline (30 µg). Susceptibility to vancomycin was detected by broth micro dilution method (CLSI 2018). *Staphylococcus aureus* ATCC® 25923 was used as negative control.

Biofilm formation assay

Biofilm production ability of the isolates was determined by microtitre plate assay as described by Vestby *et al.* (2009). The optical density (O.D.) of each well was measured at 590 nm using automated plate reader (BioRad, Irvine, CA, USA). Bacterial biofilms were classified based on OD cut-off OD_c (OD of negative control). The OD_c was defined as three standard deviations from the OD mean of the negative control. No biofilm formation was OD < OD_c; a weak biofilm former was OD_c < OD < (2 × OD_c); a moderate was biofilm former (2 × OD_c) < OD < (4 × OD_c); and a strong biofilm former was (4 × OD_c) < OD.

PCR based detection of *mec* genes

All the staphylococci were screened for *mecA* (Lu *et al.* 2016), *mecB* (Becker *et al.* 2018) and *mecC* (Bhattacharyya *et al.* 2016) genes by specific PCR assay using specific oligonucleotide primers using the method described earlier.

PCR based detection of exfoliative toxin genes

Multiplex PCR (mPCR) was carried out for detection of the exfoliative toxin genes (Set 1: *exhA*, *exhB*, *exhC*, *exhD*; Set 2: *etA*, *etB*, *etD*; Set 3: *shetA*, *shetB*) as per the methods described previously (Andersen and Ahrens 2004; Ruzikova *et al.* 2005). The *tsst-1* gene was detected separately by PCR as per the method described earlier (Monday and Bohach 1999).

PCR based screening for biofilm associated genes

All the phenotypically confirmed biofilm producing *Staphylococcus* isolates were subjected for detection of selected biofilm producing genes (*icaA* and *icaD*) as per the method described earlier (Satorres and Alcaraz 2007; Dhanawade *et al.* 2010).

Cloning and sequencing of the exfoliative toxin genes

Extracted PCR products include one *exhB* (*S. hyicus*), two *shetA* (*S. hyicus*), one *tsst1* (*S. aureus*), one *icaD* (*S. hyicus*), one *mecA* (*S. lentus*) and one *icaD* (*S. aureus*) to represent all the genes and all the bacterial species were purified and cloned in TA cloning vector and sequenced using automated sequencer Excel Applied Biosystem 3730 (USA) at DNA sequencing facility, Department of Biochemistry, University of Delhi, South Campus, New Delhi. Data were analysed using MegAlign program and phylogenetic as well as bootstrap analyses were done using neighbour joining and seqboot programs. Partial

nucleotide sequences of isolates have been deposited in NCBI GenBank (accession nos MF399267, MF383339, MF399264, MF399265, MF399266, MF399268, MF383340).

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Conflict of Interest

The authors certify that there is no conflict of interest in publication of the article.

Ethical statement

Samples were collected from the clinically infected animals without any invasive techniques with the due permission of Institutional Animal Ethics Committee, CVSc&AH, CAU, Aizawl, Mizoram.

Author contributions

SK: Collection of samples, processing for isolation and identification of bacteria, AST, PCR assays. PR: Genesis of concept of the work and preparation of the manuscript. TKD: Detection of virulence and ESBLs genes by PCR assays, editing of manuscript. PKS: AMR sensitivity assay and data analysis. SC: Sample collection and processing for isolation and identification. NNB: Preparation of manuscript and editing. AS: Preparation of manuscript and editing. All authors read and approved the final manuscript.

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